

Rogério Almeida Gouvêa

Ph.D. in Materials Science
M.Sc. in Computational Modelling

Fields: Materials Science, Computational Materials Science, Artificial Intelligence, Solid State Physics.
Skills: Machine Learning, Ab-initio calculations, Agentic Frameworks, High-throughput screening and experiments.

Address: Avenue de L'Equerre, 21, bte 104. Ottignies-Louvain-la-Neuve, 1348.
Phone: *+32 (0) 494 62 83 78*
Email: *rogeriog.em@gmail.com*

Links:
0000-0001-6837-6264
rogeriog
Rogério Gouvêa

Summary

I hold a PhD in Materials Science with a specialization in materials informatics, where I integrate first-principles simulations and machine learning to drive the development of materials for solar cells and LEDs. My career in computational materials science began with a Bachelor's internship at the University of Toronto, followed by a Master's where I gained substantial programming experience in electromagnetic simulations for solar cells.

During my PhD, I researched lead-free perovskites for solar cell and LED applications and enhanced the MODNet machine learning package during an internship at UCLouvain, significantly boosting its benchmark performance with a novel featurizer. This optimized model was instrumental in discovering new perovskite materials in my thesis.

Following my PhD, I completed a seven-month postdoctoral fellowship at UFRGS, where I focused on the realization of self-driving laboratories. During this period, I worked extensively with automated robotic platforms, integrating them with machine learning techniques to accelerate the discovery and solution-based synthesis of optoelectronic materials. Driven by the advancements in Large Language Models (LLMs), I integrated these generative tools to steer experimental design and refine research topics, bridging the gap between high-level reasoning and experimental execution.

Currently, as a Postdoctoral Researcher at UCLouvain, I am advancing the group's computational infrastructure by leading the development of several high-impact frameworks. I developed MatterVial, a hybrid framework for enhanced accuracy and explainable AI, and created VibroML to automate structural remediation using machine-learned potentials. Additionally, I co-developed MEIDNet, a multimodal generative AI framework for inverse materials design. My expertise extends to architecting custom GNNs and, more recently, integrating autonomous agentic frameworks into high-throughput workflows to realize the vision of self-driving laboratories. I am passionate about scientific innovation through robust software engineering, excel in solving complex research challenges, and possess a proven track record in team leadership and collaboration.

Languages

English ••••••
Spanish ••••••
French ••••••
Portuguese NATIVE

Programming

Python ••••••
JavaScript ••••••
Cpp / Java / Fortran / Julia ••••••
LLM Prompt engineering ••••••

Technical Skills

HPC: »Linux (bash) »Slurm »CUDA »Git/Github »CI/CD »Docker »UV

Data and visualization:
» SQL »DuckDB »MongoDB »HDF5 » Polars »Pandas »Plotly »Streamlit

Machine learning & AI:
»TensorFlow »PyTorch »Scikit-learn »HuggingFace »W&B »TensorBoard »Interpretability tools: »SHAP »LIME »Agentic frameworks (LangChain, LiteLLM, OpenRouter)

Materials Informatics:
»ASE/Pymatgen »MatMiner »APIs: OPTIMADE, Materials Project »MODNet »SISSO »MLIPs »GNN development »Atomate2

Ab-initio: »Quantum ESPRESSO »CRYSTAL »VASP »Abinit

EDUCATION / PAST EXPERIENCES

Feb 2025 - Present — *Louvain-la-neuve, Belgium*

Postdoctoral Research (MODL - UCLouvain)
P.I.: Gian-Marco Rignanese Funding: WEL-T Research Institute

May 2024 – Feb 2025 — *Porto Alegre, Brazil*

Postdoctoral Research (UFRGS)
P.I.: Marcos José Leite Santos Funding: CNPq Fellowship

- Utilizing machine learning techniques in conjunction with automated robotic platforms, we could accelerate the discovery of optoelectronic materials through solution-based synthesis. Also, conducting research on materials science by leveraging large language models within agentic framework to refine research topics and design high-throughput experiments.

May 2019 – May 2024 — *Porto Alegre, Brazil*

PhD in Materials Science (PPGCiMat - UFRGS)
Advisor: Marcos José Leite Santos Funding: CNPq PhD Scholarship

- Thesis: *Exploring $Cs_3Sb_2X_9$-type perovskites (X=Cl, Br, I) for optoelectronic applications: A theoretical investigation using density functional theory (DFT) and machine learning.*
- Ab-initio simulations supporting the development of lead-free optoelectronic materials for solar cells and LEDs. Machine learning models were used to screen other candidate materials in this family. Collaboration works include ab-initio simulations for luminescent quantum dots, and machine learning models for experimental design.

Oct 2022 – Mar 2023 — *Louvain-la-neuve, Belgium*

PhD exchange period (MODL - UCLouvain)
Co-advisor: Gian-Marco Rignanese Funding: PrInt/ CAPES PhD scholarship

- Enhanced MODNet (Materials Optimal Descriptor Network): Integrated autoencoders and graph neural networks to accelerate the featurization process and incorporate additional features, resulting in improved accuracy.
- Developed Screening Method: Utilized graph-based models CHGNet and MODNet to identify new <111>-type lead-free perovskites with low band gaps through mixed composition analysis.

Mar 2016 – Mar 2018 — *Rio Grande, Brazil*

Masters in Computational Modelling (PPGMC-FURG)
Advisor: Jeferson Ávila de Sousa Funding: CAPES Masters scholarship

- Master Thesis: *Geometric optimization of solar cells using the Constructal Design method.*
- Electromagnetic Simulation: Developed simulations for solar cells with textures generated by the automata method based on Constructal Design Theory.

Mar 2016 – Jun 2019 — *Pelotas, Brazil*

Bachelor in Physics (UFPel) (Incomplete)

- Completed advanced coursework in solid state physics, quantum mechanics II, and statistical mechanics, building a stronger foundation for theoretical and applied physics.

Mar 2010 – Dez 2015 — *Pelotas, Brazil*

Bachelor in Materials Engineering (UFPel)
Advisor: Mário Lúcio Moreira Funding: PIBITI/CNPq and PIBIC/CNPq

- Undergraduate Thesis: *First Principles Calculations for Understanding Optical, Structural, and Electronic Properties: Luminescence of BaZrO3 and ZnO/ZnS Core-Shell System.*
- Experimental Work: Synthesized refractory ceramics using biotemplates.
- Theoretical Simulations: Performed Density Functional Theory (DFT) calculations to investigate luminescent properties of materials and semiconductor band alignment.

Set 2012 – Set 2013 — *Toronto, Canada*

Sandwich period at University of Toronto
Advisor: Chandra Veer Singh Funding: Science without Borders/CNPq

- Work on theoretical calculations using Density Functional Theory (DFT) for photocatalytic materials, namely Li-doped TiO_2 for H_2 production.

PREPRINTS

1. **GOUVÊA, R. A.**; **RIGNANESE, G. M.** (2026). VibroML: an automated toolkit for high-throughput vibrational analysis and dynamic instability remediation of crystalline materials using machine-learned potentials. *arXiv preprint*, **arXiv: 2604.27685**. doi
2. **BAKHSH, M.**; **TRINQUET, V.**; **GOUVÊA, R. A.**; **RIGNANESE, G. M.**; **PONCÉ, S.** (2026). From Symmetry to Stability: Structural and Electronic Transformation in CsKInI. *arXiv preprint*, **arXiv:2602.06893**. doi
3. **BABU, A.**; **GOUVÊA, R. A.**; **VANDERGHEYNST, P.**; **RIGNANESE, G. M.** (2026). MEIDNet: Multimodal generative AI framework for inverse materials design. *arXiv preprint*, **arXiv:2601.22009**. doi

PEER-REVIEWED PUBLICATIONS

1. **GOUVÊA, R. A.**; **DE BREUCK, P.-P.**; **PRETTO, T.**; **RIGNANESE, G. M.**; **SANTOS, M. J. L.** (2026). Combining feature-based approaches with graph neural networks and symbolic regression for synergistic performance and interpretability. *npj Comput Mater* **12**, 67 (2026). doi
2. **EXNER, A.†**; **GOUVÊA, R. A.†**; **KÖCHE, A.**; **KHAN, S.**; **SANTOS, J. F. L.**; **SANTOS, M. J. L.** (2024). Doping effects on the optoelectronic properties and the stability of $Cs_3Sb_2I_9$: Density functional theory insights on photovoltaics and light-emitting devices. *Journal of Science - Advanced Materials and Devices*, **100700**, 100700. doi
3. **GOUVÊA, R. A.**; **MOREIRA, M. L.**; **SINGH, C. V.**; **SANTOS, M. J. L.** (2024). Lead-free cesium antimony halide perovskites: Halide alloying, surfaces, interfaces, and clusters. *Journal of Materials Science*, **59**, 142-160. doi
4. **PRETTO, T.**; **BAUM, F.**; **GOUVÊA, R. A.**; **BROLO, A. G.**; **SANTOS, M. J.** (2024). Optimizing the synthesis parameters of double perovskites with machine learning using a multioutput regression model. *Journal of Physical Chemistry C*, **128**, 7041-7052. doi
5. **BAUM, F.**; **PRETTO, T.**; **GOUVÊA, R. A.**; **SANTOS, M.** (2022). Design of experiments and theoretical investigation for photoluminescence optimization of copper aluminum sulfide nanoparticles through controlling crystalline defects. *Crystal Growth & Design*, **2022**, 05. doi
6. **GOUVÊA, R. A.**; **MOREIRA, M. L.**; **SOUZA, J. A.** (2019). Evolutionary design algorithm for optimal light trapping in solar cells. *Journal of Applied Physics*, **125**, 043105. doi
7. **FLORES, E. M.†**; **GOUVÊA, R. A.†**; **PIOTROWSKI, M. J.**; **MOREIRA, M. L.** (2018). Band alignment and charge transfer predictions of ZnO/ZnX (X = S, Se or Te) interfaces applied to solar cells: A PBE+U theoretical study. *Physical Chemistry Chemical Physics*, **20**, 4953-4961. doi
8. **GOUVÊA, R. A.**; **DOS SANTOS, E. D.**; **SOUZA, J. A.** (2017). Otimização construtal de uma cavidade multi-bifurcada para dissipação de calor. *Scientia Plena*, **13**, 049914-1. doi
9. **GOUVÊA, R. A.**; **FLORES, E. M.**; **CAVA, S. D. S.**; **MOREIRA, M. L.** (2016). A first-principles investigation on the luminescence emissions of $BaZrO_3$ obtained by microwave-assisted hydrothermal method. *Journal of Luminescence*, **180**, 73-80. doi
10. **FLORES, E. M.**; **RAUBACH, C. W.**; **GOUVÊA, R. A.**; **LONGO, E.**; **CAVA, S.**; **MOREIRA, M. L.** (2016). Optical and structural investigation of ZnO@ZnS core-shell nanostructures. *Materials Chemistry and Physics*, **173**, 347-354. doi
11. **DELBRÜCKE, T.**; **GOUVÊA, R. A.**; **RAUBACH, C. W.**; **LONGO, E.**; **SOUSA, V. C.**; **TEBCHERANIC, S. M.**; **KUBASKI, E. T.**; **MOREIRA, M. L.**; **JURADO, J. R.**; **CAVA, S.** (2015). Microstructure and thermal conductivity of porous Al_2O_3-ZrO_2 ceramics. *Materials Science Forum (Online)*, **820**, 268-273. doi
12. **DELBRÜCKE, T.**; **GOUVÊA, R. A.**; **MOREIRA, M. L.**; **RAUBACH, C. W.**; **VARELA, J. A.**; **LONGO, E.**; **GONÇALVES, M. R. F.**; **CAVA, S.** (2013). Sintering of porous alumina obtained by biotemplate fibers for low thermal conductivity applications. *Journal of the European Ceramic Society*, **33**, 1087-1092. doi